

ECON 351 · Chapter 9 Practice Set

12 questions total

Set 7: Insurance Problems (5)

Set 8: Risk Attitudes (4)

Set 9: Certainty Equivalent & Risk Premium (2)

Set 10: Diversification (1)

Insurance Problems

5 questions

Question 1 of 5 (Practice Q13)

Insurance #1 · HW2 Ex 3

Exercise 3 - Insurance

Tom has the following utility from income:

$$u(I) = \sqrt{x}$$

He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house. In this case, the house value becomes \$672,400. AAA offers to fully insure the house for a price P .

- a) What is the maximum insurance premium P that Tom is willing to pay?
- b) How much is the expected loss? (also known as the fair insurance premium)
- c) How much is the risk premium? (difference between the max.premium the consumer is willing to pay and the expected loss)

Question 2 of 5 (Practice Q14)

Insurance #2 · HW2 Ex 3b

Exercise 3 - Insurance

Tom has the following utility from income:

$$u(I) = \sqrt{x}$$

He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house. In this case, the house value becomes \$672,400. AAA offers to fully insure the house for a price P .

- a) What is the maximum insurance premium P that Tom is willing to pay?
- b) How much is the expected loss? (also known as the fair insurance premium)
- c) How much is the risk premium? (difference between the max.premium the consumer is willing to pay and the expected loss)

Question 3 of 5 (Practice Q15)

Insurance #3 · Sample Q18

Question 16:

Using the information from the previous question, how much is the fair insurance premium?

- (A) 174,000
- (B) 72,000
- (C) 130,500
- (D) 43,500
- (E) 168,100

Question 17:

Using the information from Question 15, find Tom's associated risk premium:

- (A) 1,200
- (B) 2,100
- (C) 1,550
- (D) 2,000
- (E) 1,875

Question 18:

Sherman has the following utility from wealth:

$$u(x) = \sqrt{x}.$$

Wealth is the sum of cash and the value of his car. He currently has a Ferrari worth \$230,900 and \$50,000 in cash (risk free). With probability 25%, a paparazzi will cause \$158,400 in damages to the car (in this case, the car value becomes \$72,500). With probability 75% nothing happens, and the car keeps its original value. AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that Sherman is willing to pay? What is the associated risk premium?

Question 19:

Joe's utility from income is given by

$$u(I) = 8\sqrt{I}.$$

He has a house in LA County. With probability $1/4$, the rain will cause a mudslide and damage his house, decreasing its value to \$250,000. With probability $3/4$, no mudslide occurs and the house retains its original \$1,000,000 value. AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Joe is willing to pay? What is the associated risk premium?

Question 20:

Marcelo has the following utility from income: $u(I) = \sqrt{I}$. He has a \$10,000 car. With probability 25% he will crash his car, and will have to pay \$6,348 to the repair shop to fix it. With probability 75% no crash occurs, and he does not have to repair the car. Progressive Insurance offers to partially insure the car for a price $P=124$. That is, Marcelo pays insurance premium $P=124$, and in case of a crash, Progressive will only pay for half of the repair cost (in case of a crash, Progressive will only pay \$3,174 to Marcelo). If Marcelo buys this partial insurance plan, his expected utility will be:

- (A) 94
- (B) 98
- (C) 96
- (D) 95
- (E) 97

Question 4 of 5 (Practice Q16)

Insurance #4 · Mock Q27

Question 27:

[Show your work] A consumer has \$1,000 in cash. With probability 90% nothing happens (he does not get sick) and he is able to keep all his money. With probability 10% he gets sick and must go to the doctor. The cost of going to the doctor is \$500.

An insurance company offers the following health insurance plan. The consumer must pay upfront a premium $p = \$100$. If the consumer gets sick, the insurance company pays the doctor, but the consumer must pay a \$275 copay.

The consumer's utility from income is given by $u(I) = \sqrt{I}$. Compute the expected utility of the consumer if he purchases this insurance.

- (A) 29.5
- (B) 30.7
- (C) 28.6
- (D) 30
- (E) 29

Question 5 of 5 (Practice Q17)

Insurance #5 · Mock Q25-26

Question 25:

[Show your work] A consumer has the following utility from income: $u(I) = \sqrt{I}$. He bought a car for \$10,000. With probability 75%, a crash will cause \$3,600 in damages to the car (in this case, the car's value becomes \$6,400). AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that the consumer is willing to pay?

- (A) 2,725
- (B) 2,850
- (C) 2,750
- (D) 2,825
- (E) 2,775

Question 26:

[Show your work] Using the information from the previous question, compute the associated risk premium.

- (A) 75
- (B) 125
- (C) 50
- (D) 150
- (E) 25

Risk Attitudes

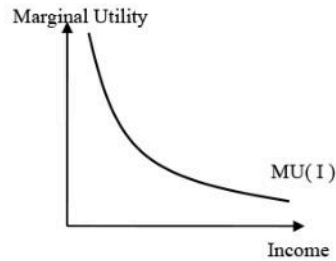
4 questions

Question 1 of 4 (Practice Q18)

Risk Attitudes #1 · HW2 Ex 2

Exercise 2 - Attitudes towards risk and fair bets

The following graph represents Mike's **marginal utility** of income (not the utility but the marginal utility):



We know that Mike's income I is greater than \$20, but we do not know his exact income. Alejandro offers Mike the following bet: Alejandro will roll a 6 sided dice (all numbers equally likely). If it is 1 or 2, Mike pays Alejandro \$20. Otherwise, if it is 3, 4, 5 or 6, Alejandro pays Mike \$10.

a) Is this a fair bet?

b) Will Mike accept or reject the bet? Does your answer depend on Mike's initial wealth? (assuming he can always pay the \$20 if he loses)

Question 2 of 4 (Practice Q19)

Risk Attitudes #2 · Sample Q6

- a) Compute the expected value of each investment option.
- b) Compute the expected utility of each investment option. Find the best (and worse) investment option for Jim.

Question 3:

Pedro Carroll has the following utility from income: $u(I) = \sqrt{I}$. Currently, he has an income $I = \$169$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Seahawks to win the championship game. If he bets all his money and the Seahawks win, then he ends up with an income $I = \$400$. If he bets all his money and the Seahawks lose, then he ends up with nothing, $I = \$0$. If he does not bet, he keeps his \$169 for sure.

The Los Angeles Seahawks will win the championship game with probability P , and will lose the game with probability $1-P$.

Compute the minimum winning probability P such that Pedro would be willing to bet all his money on the Seahawks.

- (A) $P = 50\%$
- (B) $P = 65\%$
- (C) $P = 70\%$
- (D) $P = 55\%$
- (E) $P = 60\%$

Question 4:

Cam has the following utility from income: $u(I) = \sqrt{I}$. Currently he has an income $I = \$36$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Panthers to win the championship game. If he bets all his money and

the Panthers win, then he ends up with an income I_{Win} . If he bets all his money and the Panthers lose, then he ends up with nothing, $I_{Lose} = \$0$. If he does not bet, he keeps his \$36 for sure.

The Los Angeles Panthers will win the championship game with probability 75%, and will lose the game with probability 25%.

Compute the minimum winning prize I_{Win} such that Cam would be willing to bet all his money on the Panthers.

- (A) $I_{Win} = \$36$
- (B) $I_{Win} = \$48$
- (C) $I_{Win} = \$56$
- (D) $I_{Win} = \$64$
- (E) $I_{Win} = \$72$

**Question 6:**

Anna's utility of income function is $U(I) = I$, while Elsa's utility of income function is $U(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

Question 3 of 4 (Practice Q20)

Risk Attitudes #3 · Sample Q8

- (A) risk neutral, risk loving.
 (B) risk averse, risk averse.
 (C) risk loving, risk loving.
 (D) risk averse, risk loving.
 (E) risk loving, risk averse.

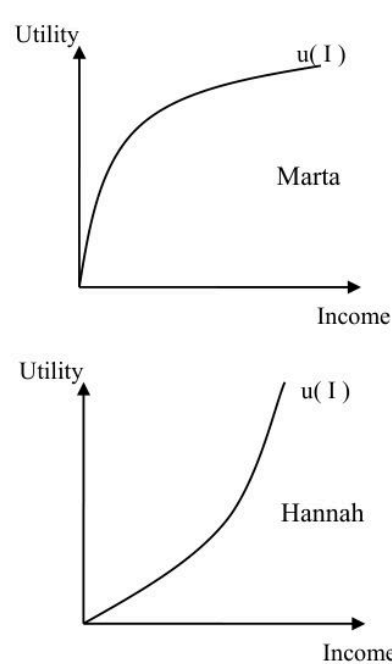
Question 7:

Anna's marginal utility of income function is $MU(I) = I$, while Elsa's marginal utility of income function is $MU(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
 (B) risk averse, risk averse.
 (C) risk loving, risk loving.
 (D) risk averse, risk loving.
 (E) risk loving, risk averse.

Question 8:

The following graphs represent the utility function of consumers Marta and Hannah:

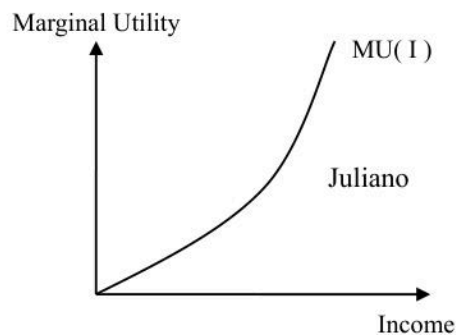
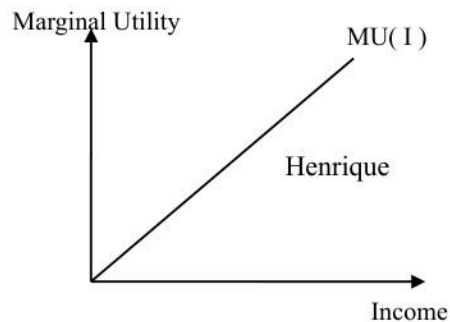


Using the information from the graphs,

- (A) Both Marta and Hannah are risk loving
 (B) Both Marta and Hannah are risk averse
 (C) Marta is risk averse; Hannah is risk loving
 (D) Marta is risk loving; Hannah is risk averse
 (E) Not enough information from the graphs

Question 9:

The following graphs represent the marginal utility of income functions of consumers Henrique and Juliano.



Using the information from the graphs, Henrique is _____ and Juliano is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 10:

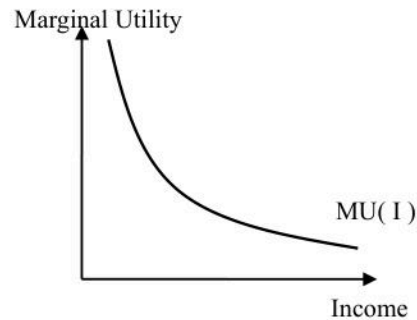
Mike has a constant marginal utility of income. We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough

- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 11:

The following graph represents Mike's marginal utility of income:



We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough
- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 4 of 4 (Practice Q21)

Risk Attitudes #4 · Mock Q28

Question 28:

Firmino is a big fan of a soccer team called The Reds. Everyone knows that there is a 75% chance that the The Reds will win the next tournament, and a 25% chance that they will lose. Firmino offers the following bet to each one of his friends: if The Reds win, then the friend must pay \$1 to Firmino. If The Reds lose, then Firmino will pay \$3 to the friend.

The friends Mohamed and Sadio (who are very rich) are each considering whether to accept the bet. Mohamed's utility of income function is

$$U(I) = I^{0.7}$$

while Sadio's utility of income function is

$$U(I) = I^{2.3}$$

Therefore,

- (A) Both Mohamed and Sadio will reject the bet
- (B) Both Mohamed and Sadio will accept the bet
- (C) Mohamed will accept and Sadio will reject the bet
- (D) Mohamed will reject and Sadio will accept the bet
- (E) Both Mohamed and Sadio are indifferent between accepting and rejecting the bet

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Question 30:

Mirabel's marginal utility of income function is $MU(I) = 9$, while Bruno's marginal utility of income function is $MU(I) = \frac{3}{I}$. Therefore, Mirabel is _____ and Bruno is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk neutral, risk averse.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Certainty Equivalent & Risk Premium

2 questions

Question 1 of 2 (Practice Q22)

CE & RP #1 · HW2 Ex 1

Exercise 1 - Risk Aversion and Certainty Equivalent

Jackie owns shares of a company. With probability 80%, the value of the investment will go up to $x = \$200$, and with probability 20% the value of the investment will go down to $x = \$50$. Jackie's utility from income is given by the function

$$u(x) = \sqrt{x}$$

- a) Compute the expected value of the investment and Jackie's expected utility from holding this investment.
- b) Compute the certainty equivalent of this investment. In other words, the certain amount that makes Jackie indifferent between holding the investment or selling it for that certain amount (the certain amount provides the same expected utility as the investment).
- c) Compute the risk premium associated with this investment.

Question 2 of 2 (Practice Q23)

CE & RP #2 · Sample Q12

Question 12:

A consumer owns shares of a company. With probability 80% the value of the investment will go up to $x = \$144$, and with probability 20% the value of the investment will go down to $x = \$81$. The consumer's utility from income is given by the function

$$u(x) = \sqrt{x}.$$

- Compute the consumer's expected utility from holding this investment.
- Compute the expected value of this investment.
- Compute the certainty equivalent of this investment.
- Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- $P = 94$
- $P = 130$
- $P = 121$
- $P = 104$
- $P = 100$

Question 14:

Using the information from the previous question, compute the associated risk premium.

- 3
- 1
- 7
- 10
- 4

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- 23,010
- 45,375
- 74,000
- 32,375
- 15,600

Diversification

1 question

Question 1 of 1 (Practice Q24)

Diversification · Sample Q22

Question 21:

Arthur has the following utility from income:

$$u(I) = \sqrt{I}.$$

He has a collection of rare comic books that is worth \$10,000. With probability 50% a silverfish invasion will destroy all his books, bringing the value of his collection to zero. With probability 50% nothing occurs, and his collection keeps its full value. A company called Insect Invasion Insurance (III) offers to partially insure the collection for a price $P = \$1,900$. That is, Arthur first pays the insurance premium $P = \$1,900$; and then, in case of a silverfish invasion, III will send to Arthur a \$2,044 check. If Arthur buys this partial insurance plan, then his expected utility will be _____. Therefore, Arthur _____.

[Be careful with your computation. Keep as many decimal places as possible along all the steps of the problem. Once you are done with all computations, round your final answer to the nearest integer. It should be clear which one.]

- (A) 53 ; should buy the insurance
- (B) 51 ; should buy the insurance
- (C) 51 ; should not buy the insurance
- (D) 53 ; should not buy the insurance
- (E) None of the above, because Arthur is indifferent between buying the insurance or not.

Question 22:

Feng has all her wealth invested in Company A and in Company B. These are agricultural companies, and their value can go up or down depending on the amount of rainfall this season. There is a 50% chance that it

will rain a lot, in which case the value of her investment in Company A will go up to \$100 while her investment in Company B will go up to \$90. There is a 50% chance of a drought, in which case the value of her investment in Company A will go down to \$80, while her investment in Company B will go down to \$70.

Nan has her wealth invested in Company C and in Company D. Company C is an airline company, while Company D is an oil company. Their value can go up or down depending on the international price of oil. There is a 50% chance that the price of oil will go up, in which case her investment in Company D will go up to \$100 while her investment in Company C will go down to \$70. There is a 50% chance that the price of oil will go down, in which case the value of her investment in Company D will go down to \$80, while her investment in Company C will go up to \$90.

- a) Are the values of Company A and Company B positively or negatively correlated?
- b) Are the values of Company C and Company D positively or negatively correlated?
- c) Who has the riskier investment portfolio, Feng or Nan?

Question 23:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$120	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

Question 24:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$102	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

b) _____
